

MARS

Malware Analysis & Reverse-engineering System

ANALYSIS ID	MARS-20260620054031-DA6FD8
TARGET FILE	N/A
GENERATED	2026-06-20

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Executive Summary

CLEAN

0.0 / 10.0
Severity Score

Analysis Summary

No significant threat indicators were identified during static or dynamic analysis. The sample appears clean based on available evidence; manual review is still recommended for high-value targets.

Sample Identification

Filename	N/A <i>Original filename of the submitted sample as read from disk.</i>
MD5	0c411b04b00fb818b42ba31a0d5e1751 <i>MD5 digest of the file — used for fast de-duplication checks.</i>
SHA256	e7dea3de7e8ca278175fbea753cd56f8c76661909df630f9ee08c8c7b1a51878 <i>SHA-256 cryptographic hash — primary unique identifier for the sample.</i>
Submission Timestamp	N/A <i>UTC date and time at which the file was submitted to the pipeline.</i>

Indicators of Compromise (IoC Summary)

Consolidated, copy-pasteable indicators extracted from all analysis phases. Use these directly for threat hunting, firewall rules, and SIEM ingestion.

Cryptographic Hashes

Type	Value
MD5	0c411b04b00fb818b42ba31a0d5e1751
SHA1	8fc4a3868898c2b50d85e81b888d6bf8f5698cd8
SHA256	e7dea3de7e8ca278175fbea753cd56f8c76661909df630f9ee08c8c7b1a51878

MITRE ATT&CK Mapping

Observed behaviours cross-referenced with the MITRE ATT&CK; framework. Each row maps a detected indicator to its technique ID, name, and tactic phase.

No MITRE ATT&CK; techniques could be mapped from the available evidence.

1. File Intake Summary

Metadata captured at ingestion time. Hash values are computed immediately on submission before any analysis step touches the file.

Analysis ID	MARS-20260620054031-DA6FD8 <i>Unique identifier auto-generated for this analysis session.</i>
Original File Name	reverse_shell.exe
File Size (Bytes)	40850
Extension	.exe <i>File extension used to determine the analysis pipeline branch.</i>
Creation Timestamp	2026-06-20 05:19:36
Magic Bytes (Hex)	4D5A9000
MIME/Format Guess	application/x-msdownload (PE Executable)
MD5	0c411b04b00fb818b42ba31a0d5e1751 <i>MD5 digest of the file — used for fast de-duplication checks.</i>
SHA1	8fc4a3868898c2b50d85e81b888d6bf8f5698cd8 <i>SHA-1 digest of the file.</i>
SHA256	e7dea3de7e8ca278175fbea753cd56f8c76661909df630f9ee08c8c7b1a51878 <i>SHA-256 cryptographic hash — primary unique identifier for the sample.</i>
SHA512	d9a7f9309e006f5e9cec572562d992d5a1812fd6b2df118505c511d9494a9889

2. Package & Archive Unpacking

Not applicable — the submitted file is not a ZIP archive or MSI package.

3. Deep Static Analysis

PE file inspection performed without executing the sample. Covers header metadata, security mitigations, section entropy, IAT analysis, manifest parsing, string extraction, and YARA signature matching.

Target: reverse_shell.exe

PE Headers

Machine Architecture	0x14c <i>Target CPU architecture encoded in the COFF File Header (e.g. 0x14c = x86, 0x8664 = x64).</i>
Compile Timestamp	2026-04-04 17:54:16 <i>Date/time the binary was linked. Stored in the PE File Header — trivially forgeable.</i>

DOS Header Offset (e_lfanew)	0x80 <i>Byte offset of the PE signature from file start. Abnormal values may indicate header manipulation.</i>
Address of Entry Point	0x12e0 <i>Relative virtual address (RVA) of the first instruction executed. Unusual RVAs suggest packing.</i>
Image Base	0x400000 <i>Preferred virtual address where the image is loaded. 0x400000 is the traditional default.</i>
Number of Sections	13 <i>Count of PE sections. Values outside 3-10 are often associated with packers or malformed binaries.</i>

Mitigations

DEP / NX Bit	Disabled <i>Data Execution Prevention — marks data pages non-executable to block shellcode.</i>
ASLR	Disabled <i>Address Space Layout Randomisation — randomises load addresses to hinder ROP chains.</i>
CFG (Control Flow Guard)	Disabled <i>Microsoft's compile-time + runtime control-flow integrity check on indirect calls.</i>
RFG (Return Flow Guard)	Disabled <i>Return-flow integrity check; guards return addresses against ROP corruption.</i>
SafeSEH	Disabled <i>Structured Exception Handler protection — validates handlers against a compile-time list.</i>
Stack Canaries (/GS)	Disabled <i>Compiler-inserted stack cookie; detects stack-buffer overflows before return.</i>
Hardware Protection (CET/Shadow Stack)	Disabled <i>Intel CET Shadow Stack — hardware-enforced return-address integrity (processor support required).</i>
Force Integrity	Disabled <i>Requires a valid Authenticode code-signing signature before the loader will map the image.</i>
Isolation / AppContainer	Disabled <i>Binary is declared to run inside an AppContainer with restricted privilege and syscall surface.</i>

Sections

Section: .text	Perms: R-E Entropy: 6.19
Section: .data	Perms: RW- Entropy: 0.22
Section: .rdata	Perms: R-- Entropy: 4.79
Section: /4	Perms: R-- Entropy: 4.73
Section: .bss	Perms: RW- Entropy: 0.00
Section: .idata	Perms: RW- Entropy: 4.57
Section: .CRT	Perms: RW- Entropy: 0.11
Section: .tls	Perms: RW- Entropy: 0.22
Section: /14	Perms: R-- Entropy: 0.22
Section: /29	Perms: R-- Entropy: 5.77

Section: /41	Perms: R-- Entropy: 3.04
Section: /55	Perms: R-- Entropy: 4.32
Section: /67	Perms: R-- Entropy: 0.67

Suspicious Imports

Tracked APIs Found	0 <i>Number of monitored Win32 API imports found in the Import Address Table (IAT).</i>
APIs	None detected <i>Comma-separated list of suspicious API names. Cross-reference with MITRE ATT&CK; technique IDs.</i>

Manifest Data

Manifest Status	Not Found <i>Whether a valid XML application manifest was found and parsed inside the PE resources.</i>
Requested Execution Level	Unknown <i>Privilege level declared in the manifest: asInvoker highestAvailable requireAdministrator.</i>

Strings Analytics

IPv4	1
IPv6	0
URL	0
Registry	0 <i>Windows Registry key reads, writes, and deletions captured during execution.</i>
Email	0
Password-Like	0

Extracted Artifacts

IPv4	['10.9.1.6']
IPv6	[]
URL	[]
Registry	[] <i>Windows Registry key reads, writes, and deletions captured during execution.</i>
Email	[]
Password-Like	[]

YARA Signatures

Hits	0 <i>Total number of YARA rule signatures that matched against the sample.</i>
Matched Rules	Clean <i>YARA rule names that fired. Each name encodes a threat family or behavioural pattern.</i>

4. Dynamic Sandbox Analysis

Behavioural telemetry captured during live detonation in the VMware sandbox. The in-guest agent instruments filesystem, registry, network, process, memory, and hardware subsystems in real time.

No dynamic analysis results — the dynamic module was either skipped or the VMware sandbox is not configured.